

Entropy

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Following the well-known paper of C.E.Shannon [1], we present some fundamental properties of entropy in information theory.

Definition 1. *Given a set of possible events with probability occurrence $p_1, p_2, \dots, p_n$. The entropy H is defined as*

$$H = -K \sum_{i=1}^n p_i \log(p_i), \quad (1)$$

where K is a positive constant.

Denote $H = H(p_1, p_2, \dots, p_n)$

With the formula in the definition, we have some properties regarding to entropy (mentioning in Wikipedia [2]).

Proposition 1. *H is satisfied some properties.*

- **Continuity:** *H is continuous with respect to all p_i .*
- **Symmetry:** *The value of H is unchanged for any permutations of p_i .*
- **Maximum:** *H is maximum when all probabilities p_i are equal.*
- **Increasing property:** *Under the condition of equiprobable events, the entropy increase when all outcomes increase, i.e*

$$H\left(\frac{1}{n}, \frac{1}{n}, \dots, \frac{1}{n}\right) < H\left(\frac{1}{n+1}, \frac{1}{n+1}, \dots, \frac{1}{n+1}\right).$$

- **Additivity:**

The properties of entropy for random variables X, Y is described as

Proposition 2. • **Subadditivity:** *$H(X, Y) \leq H(X) + H(Y)$ for jointly distributed random variables X, Y .*

- **Additivity:** *$H(X, Y) = H(X) + H(Y)$ if X and Y are independent.*
- **Expansibility:** *H is maximum when all probabilities p_i are equal.*

- **Symmetry:** Under the condition of equiprobable events, the entropy increase when all outcomes increase, i.e

$$H\left(\frac{1}{n}, \frac{1}{n}, \dots, \frac{1}{n}\right) < H\left(\frac{1}{n+1}, \frac{1}{n+1}, \dots, \frac{1}{n+1}\right).$$

- **Small for small probability:** $\lim_{q \rightarrow 0} H(q, 1 - q) = 0$.

References

- [1] Claude Elwood Shannon. A mathematical theory of communication. *The Bell System Technical Journal*, 27:379–423, 1948.
- [2] Wikipedia contributors. Entropy (information theory) — wikipedia, the free encyclopedia, 2026. Accessed: 2026-05-25.